ACADEMIC REPORT: MOVIE GENRE CLASSIFICATION

Predicting Movie Genres from Plot Summaries using TF-IDF & Multinomial Naive Bayes

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Course: Machine Learning & Natural Language Processing
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1. PROBLEM STATEMENT

With hundreds of thousands of motion pictures released globally, streaming platforms and digital archives require automated categorization mechanisms to organize large media catalogs, enhance search indexing, and power personalized recommendation systems. Manual genre tagging is labor-intensive, subjective, and prone to inconsistency. This project addresses the challenge of automated text classification: given an unstructured textual plot summary of a movie, build and evaluate a machine learning pipeline to accurately predict its primary cinematic genre (e.g., Action, Comedy, Drama, Horror, Romance, Sci-Fi, Thriller).

2. TEXT DATA PREPROCESSING & FEATURE EXTRACTION (TF-IDF)

Raw movie plot descriptions consist of unstructured natural language containing noise, syntactic variations, and non-informative tokens. The preprocessing pipeline applies:
a. Text Normalization: Conversion of all characters to lowercase to eliminate case sensitivity (e.g., "Alien" vs. "alien").
b. Noise Reduction: Regular-expression removal of HTML markup, URLs, numerical digits, and punctuation characters.
c. Stop Word Removal: Elimination of high-frequency English functional words (such as "the", "and", "is", "with") that carry minimal genre-specific discriminative information.
d. Tokenization & N-gram Extraction: Splitting text into individual tokens as well as bi-grams (1, 2) to preserve meaningful domain phrases (e.g., "serial killer", "deep space", "romantic dinner").
e. TF-IDF Vectorization: Term Frequency-Inverse Document Frequency (TF-IDF) transforms the variable-length text into high-dimensional numerical feature vectors:
TF-IDF(t, d, D) = TF(t, d) * log((1 + |D|) / (1 + DF(t))) + 1
This penalizes universally ubiquitous terms while rewarding words that are distinctive to specific genres. A vocabulary threshold of the top 5,000 features with sublinear TF scaling is implemented to control sparsity and prevent term saturation.

3. ALGORITHM SELECTION: WHY MULTINOMIAL NAIVE BAYES FOR TEXT?

The Multinomial Naive Bayes (MNB) classifier was selected as the optimal algorithm for this task due to several theoretical and empirical strengths in Natural Language Processing:
a. High-Dimensional Text Suitability: Text classification involves thousands of feature dimensions. Naive Bayes performs exceptionally well in high-dimensional, sparse vector spaces where complex discriminative models often overfit.
b. Multinomial Distribution Match: MNB is mathematically formulated to model discrete event counts and word frequency distributions (and by extension TF-IDF weights) occurring in text documents.
c. Computational Efficiency: Training and inference exhibit linear time complexity O(N * |V|), making it orders of magnitude faster to train and deploy than support vector machines or recurrent neural networks.

d. Laplace Smoothing (alpha = 0.5): Prevents zero-probability penalties for unseen vocabulary terms during testing, maintaining robust probabilistic calibration.

4. EXPERIMENTAL SETUP & EVALUATION METRICS

The dataset was partitioned into an 80% training set and a 20% test set using stratified sampling to preserve class distribution across genres. Model performance was evaluated using the following standard statistical metrics:

- Accuracy: Overall proportion of correct predictions across all test instances.
- Precision: Proportion of predicted instances for a genre that were truly correct, minimizing false positives.
- Recall: Sensitivity metric measuring the proportion of actual genre instances correctly identified, minimizing false negatives.
- F1-Score (Macro & Weighted): Harmonic mean of precision and recall, balancing performance across both majority and minority genre classes.
- Confusion Matrix: Inter-class confusion tracking to observe genre misclassifications.

5. ANALYSIS OF RESULTS & OBSERVATIONS

Experimental evaluation revealed the following key insights:

a. Lexical Distinctiveness: Genres with unique, highly specialized vocabularies—such as Horror ("ghost", "curse", "haunted", "demonic"), Sci-Fi ("spaceship", "alien", "planet", "future"), and Romance ("love", "wedding", "passion")—achieved the highest individual precision and recall scores.

b. Multi-Genre Overlap: Genres with broader narrative thematic scope, particularly Drama and Thriller, exhibited higher misclassification rates. Because a dramatic plot can easily overlap with crime, romance, or action themes, single-label classification faces inherent linguistic ambiguity.

c. Feature Sparsity Management: Incorporating sublinear term frequency scaling and bi-grams significantly improved discrimination between nuanced sub-genres compared to basic bag-of-words count vectorization.

6. LIMITATIONS & FUTURE SCOPE

While Multinomial Naive Bayes with TF-IDF provides a lightweight, highly interpretable, and low-latency baseline, real-world movie summaries frequently belong to multiple genres simultaneously (e.g., Sci-Fi Comedy). Future enhancements include:

1. Multi-Label Classification: Transitioning to Binary Relevance or Classifier Chains

to predict multi-genre tags per movie.

2. Contextual Transformer Embeddings: Implementing contextual language models (such as

BERT or RoBERTa) to capture semantic dependencies, polysemy, and narrative structure beyond keyword occurrence.

7. CONCLUSION

The end-to-end pipeline demonstrates that TF-IDF feature extraction combined with a Multinomial Naive Bayes classifier delivers an effective, robust, and computationally efficient solution for movie genre prediction from synopsis text, fulfilling all academic and production deployment objectives.